

STATEMENT OF WORK

Business Requirements Document (BRD)

Quality and Document Governance Report

QA/QC Report, Customer Satisfaction, Daily Document Submission Report

– Web & Mobile Application

Abstract

This SOW defines the scope, functional requirements, workflow, roles, escalation rules, and deliverables for the unified QMS System. The system consolidates three modules — the Monthly QA/QC Report, Customer Satisfaction Survey, Daily Document Submission Report— into a single digital platform. It captures the most important QAQC KPIs, customer feedback, and Daily document status providing relevant project scores and compliance visibility. It aims to improve quality score and customer satisfaction, aid the top management to compare similar projects, and enhance overall quality and document compliance.

Document Title	SOW – Business Requirements Document Quality and Document Governance Report
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1. Document Control

1.1 Revision History

Version	Date	Author	Reviewed By	Description of Changes
6.0v1.0	14-06-2026	Subbu Sankar V R	Owies Alrababah	Mentioned in Comments
v2.0	17-06-2026	Subbu Sankar V R	Owies Alrababah	
v3.0	16-07-2026	Subbu Sankar V R	Owies Alrababah	Expanded scope to unified QMS System; added Customer Satisfaction and Biweekly Document Submission Report
v4.0	06-08-2026	Subbu Sankar V R	Ihtisham Hamid	Changed Biweekly Document Submission to Daily Document Governance Log.
v5.0	17-08-2026	Subbu Sankar V R	Ihtisham Hamid	Removed Rejected Submission dashboard visual, moved Submittals & Drawings visuals to the top, added Excel template upload functionality, and changed email distribution of document submission module frequency to fortnightly.
V6.0	18-08-2026	Subbu Sankar V R		Added automated submission delay alerts, introduced 'Zero-Fill' toggle for days/months with no updates, enforced sequential submission dependency, and updated historical data fetch to include unapproved previous submissions.

2. Executive Summary

This SOW defines the business requirements for the design, development, and implementation of a unified Quality and Document Governance Report to digitize and centralize key Quality Assurance and Quality Control (QAQC) processes across Alghaz projects. The primary objective is to replace manual data collection and reporting with a structured digital platform that automates KPI reporting, improves data accuracy, and provides actionable insights through dashboards and reports.

The proposed solution consolidates multiple quality functions—including QAQC Metrics, Quality Toolbox Talks (QTBT), Material Inspection, QMS Reporting, Customer Satisfaction, Daily Document Governance Log—into a single portal with dedicated submission and approval workflows. In the initial phase, Quality representatives will manually update the required data, with future integration planned for other QAQC modules.

This BRD is intended for business stakeholders, QAQC teams, project teams and implementation teams involved in defining, developing, reviewing, and deploying the solution. It outlines the business requirements, functional

scope, and expected deliverables to ensure the successful implementation of a centralized QMS that enhances quality compliance, reporting efficiency, document traceability, and management decision-making.

3. Project Description

3.1 Project Background

- The objective of this project is to design and implement a unified Quality and Document Governance Report that digitizes and centralizes key Quality Assurance and Quality Control (QAQC) processes.
- Currently, the QAQC team prepares the monthly Quality Report manually by consolidating data from multiple reports, performing KPI calculations using Excel formulas, obtaining approvals, and distributing the report to project teams.
- This process helps monitor project quality compliance and the Project Quality Index (PQI), but it is highly dependent on manual effort.
- Due to the difficulties involved in the preparation of this report and the need for a structured digital module, this solution has been proposed

3.2 Key Pain Points & Gap in Current Process

#	Area	Pain Point / Gap	Impact
1	Data Collection	Data is being collected for each metric from different reports and requires manual counting.	Time consuming process and it has huge risk of data errors.
2	Calculations	Manual Excel formulas are used in the calculation of KPIs	Error in formula could lead to serious issues.
3	Historical Data	Past data for these metrics need to be referred manually which in turn requires re-entering.	It is always required to maintain historical versions which increases the manual documentation.
4	Approval & Distribution	The prepared report requires an approval and to be sent across projects.	Manual intervention in getting approval and distribution delays the process. Also, it's quite difficult to trace.
5	Visibility	The dashboard & report is not centralised which makes these real time KPIs as ineffective.	The non-availability of these metrics to all reduces the awareness of the need to maintain good compliance.
6	Quality Assessment	Project brief requires manual writing for each project every time.	Project brief suggestions are always dependent on the individual, which sometimes becomes ineffective due to human errors.
7	Customer Feedback	Customer satisfaction is collected informally and not captured digitally.	No structured trend data on client satisfaction; feedback is not linked to the project record.
8	Document Status	Document statuses are maintained in scattered spreadsheets	Difficult to retrieve the document status.

4. Project Scope & Deliverables

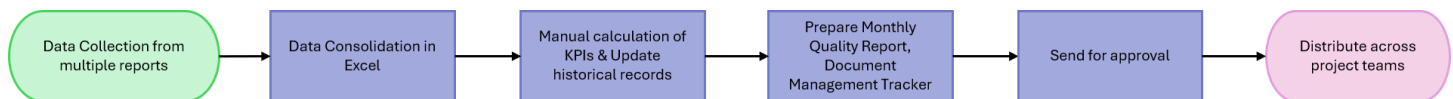
Scope	Description	Platform	Scope
Development of QAQC Metrics Module	Log the overall numbers of External/Internal NCR, RFI & RMI for the whole month with ageing analysis along with docs status.	Web & Mobile	In-Scope
Development of QTBT Module	Track the number of toolboxes talk across the projects with manhour calculation.	Web & Mobile	In-Scope
Material Inspection Module	Capture the number of inspections with respect to MIRN numbers and status of material also in addition to OSD.	Web & Mobile	In-Scope
Customer Satisfaction Survey	Capture structured client satisfaction ratings and qualitative feedback per project.	Web & Mobile	In-Scope
Daily Document Governance Log	Maintain a centralized status for the submitted submittals and drawings across the projects	Web & Mobile	In-Scope
Historical Data	Auto carry-over of previous month/day submitted figures.	Web & Mobile	In-Scope
Auto-Calculation Engine	Real-time calculation of all derived KPI metrics as per the provided formula	Web & Mobile	In-Scope
Approval Workflow	Route reports for approval with revision and comments.	Web & Mobile	In-Scope
Master Management	Configure projects, targets, users, and distribution lists for the email as per the list provided in section 9.	Web & Mobile	In-Scope
Dashboard & Report Distribution	Main Dashboard, Dashboard Report and Report to be developed as per digital team format. Auto-generate PDF and distribute to sites after the approval. Monthly Report – Monthly basis Daily document status – Fortnightly basis	Web & Mobile	In-Scope
Excel Data Upload	Functionality to download a standardized Excel template and upload data as an alternative to manual form filling.	Web	In-Scope
Data Export	Export reports and raw data in PDF and Excel formats.	Web & Mobile	In-Scope
Role-Based Access Control	Defined roles with clearly defined access for each module as per the list is provided in Section 11	Web & Mobile	In-Scope
AI Quality Brief Generator	AI-assisted draft generation for Quality Assessment Brief.	Web & Mobile	In-Scope
Integration	API-based integration with NCR, RFI, RMI capture modules and Unifier(Document Governance Report)	Web & Mobile (Phase 2)	Out Scope
User Documentation & Training Material	User manuals, training guides, and videos.	-	In-Scope
UAT Support	Support during User Acceptance Testing phase.	-	In-Scope
Post Go-Live Support	Support after Go-Live in case of any issues or bugs.	-	In-Scope

4. Business Drivers

Drivers	Description
Central QMS system	Standardized quality metrics reporting across projects aids in managing a central system.
Operational Efficiency	Manual effort in preparing a form, dashboard, report is significantly reduced and significant time is saved.
Real Time Visibility	Complete transparency on the quality KPIs offers real time visibility.
Error Minimization	Auto calculation eliminates the human error and improves data accuracy
Decision Making	Enhance decision-making through dashboards and automated reports.

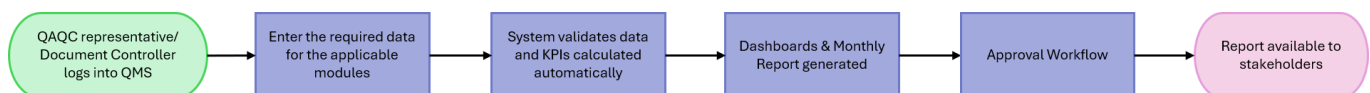
5. Current Process

- The QAQC Monthly Report is currently prepared manually by the Quality team monthly.
- Quality representatives collect the required data from multiple reports, registers, and project records, and manually consolidate it into an Excel-based reporting template.
- KPI calculations are performed using predefined formulas, after which historical data is manually referred to and updated.
- Once the report is completed, it is submitted for approval and subsequently circulated to the respective project teams.
- Similarly, Customer Satisfaction records, Document Registers are maintained separately in spreadsheets and updated manually.
- This fragmented approach results in significant manual effort, increased risk of errors, and limited visibility of quality information across projects.



6. Proposed Process

The proposed solution introduces a centralized Quality Management System (QMS) through which Quality representatives will capture quality-related information using dedicated modules, via manual UI entry or by uploading a standardized Excel template. The system will automatically calculate KPIs, generate dashboards and monthly reports, route them through the approval workflow, and make the approved reports available to project stakeholders. In future phases, the system will integrate with other QAQC modules to eliminate manual data entry



and further automate the reporting process.

This section defines all data fields, UI input types, and validation/calculation logic for each section of the QMS system

6.1 QAQC Metrics

6.1.1 Project Details Section

Field Name	UI Input Type	Validation / Calculation Logic
Project Name	Dropdown	Pulled from active Project Master list. Mandatory.
Month / Year	Month/Year Picker (MM-YYYY).	Restricts selection to valid reporting periods System must prevent duplicate submissions for same project-month combination.
Cost Centre	Auto-fill (Read-Only)	Auto-generated based on selected Project Name

6.1.2 PQP Status

Field Name	UI Input Type	Validation / Calculation Logic
PQP Status	Dropdown	Approved A, Approved B, Approved C, Under Preparation, Under Review with Client, Rejected & Others If others is selected a text box should capture the user defined status
PQP Date of Submission	Date Picker	Should be enabled for all status except Under Preparation
PQP Date of Approval	Date Picker	Should be enabled only for Approved A/B/C

6.1.3 Report Reference

Field Name	UI Input Type	Validation / Calculation Logic
Last Report Sent to Client (Ref. No)	Text field	Optional
Report Date(from)	Date Picker	Mandatory if Last Report Sent to Client is entered
Report Date (To)	Date Picker	Mandatory if Last Report Sent to Client is entered

6.1.4 Meetings Conducted

Field Name	UI Input Type	Validation / Calculation Logic
Meeting	Dropdown	Project Management Review Meeting, Internal meeting, External meeting & Project Quality Meeting
Report Date(from)	Date Picker	Mandatory if any Meeting is selected
Report Date (To)	Date Picker	Mandatory if any Meeting is selected

6.1.5 QMS Internal Audit

Field Name	UI Input Type	Validation / Calculation Logic
Is Internal Audit Conducted?	Dropdown	Yes, No
Last Internal Audit Conducted Date	Date Picker	Mandatory
Next Internal Audit Conducted Date	Date Picker	Mandatory

6.1.6 Quality Manpower

Field Name	UI Input Type	Validation / Calculation Logic
Department	Text Field	Mandatory
Nos	Numeric	Integer >0
Is client approval required?	Dropdown	Yes, No
How many approved	Numeric	Mandatory if Client Approval is required Integer >=0
How many rejected	Numeric	Mandatory if Client Approval is required Integer >=0
Total Manpower	Numeric	Summation of Nos

6.1.7 QA/QC Metric Details Section (Ext NCR, Int NCR, RFI & RMI)

Field Name	UI Input Type	Validation / Calculation Logic
Acc. till Last Month	Pre-filled (Read-Only)	Auto populated from previous month's submitted Acc. till this Month value. Defaults to 0 if no prior record exists.
Closed till Last Month	Pre-filled (Read-Only)	Auto populated from previous month's submitted Closed till this Month value. Defaults to 0 if no prior record exists.
Issued this Month	Numerical Input	Integer ≥ 0 . Mandatory. Phase 1: Manual entry. Phase 2: Auto pulled from NCR/RFI/RMI module via integration.
Closed this Month	Numerical Input	Integer ≥ 0 . Mandatory. Phase 1: Manual entry. Phase 2: Auto pulled from NCR/RFI/RMI module via integration Must not exceed (Acc. till Last Month + Issued this Month).
Acc. till this Month	Auto-Calculation (Read-Only)	= [Acc. till Last Month] + [Issued this Month]
Closed till this Month	Auto-Calculation (Read-Only)	= [Closed till Last Month] + [Closed this Month]
Closure Rate this Month	Auto-Calculation (Read-Only)	= ([Closed this Month] / [Issued this Month]) x 100. Special Rule: If both Issued and Closed = 0 and project is active, Closure Rate = 100%.
% Acc. till Last Month	Auto-Calculation (Read-Only)	= ([Closed till Last Month] / [Acc. till Last Month]) x 100. Special Rule: If both are 0 and project is active, % acc till last month = 100%.
% Acc. till this Month	Auto-Calculation (Read-Only)	= ([Closed till this Month] / [Acc. till this Month]) x 100. Special Rule: If both are 0 and project is active, % acc till this month = 100%.
Variance	Auto-Calculation (Read-Only)	= [% Acc. till this Month] - [% Acc. till Last Month]. Positive = improvement. Display with +/- indicator and colour coding.
Open NCRs (only for Ext/Int NCR)	Auto-Calculation (Read-Only)	Acc. till this Month - Acc. Closed till this Month
BU/Dept (only for Ext/Int NCR)	Dropdown	Will be provided
Ageing Bucket (only for Ext/Int NCR)	Dropdown	0-15 days, 15-45 days & >45 days
Count (only for Ext/Int NCR)	Numeric	Integer >0 The sum of count should be equal to Open NCRs

6.1.8 PQI (Project Quality Index) Calculation

PQI is a composite index derived from the four metric categories. It is entirely system-calculated and not editable.

Field Name	UI Input Type	Validation / Calculation Logic
PQI - Acc. till Last Month	Auto-Calculation (Read-Only)	= Average of (% Acc. till Last Month) across Ext NCR, Int NCR, RFI, and RMI for the same project-month.

Field Name	UI Input Type	Validation / Calculation Logic
PQI - Acc. till this Month	Auto-Calculation (Read-Only)	= Average of (% Acc. till this Month) across Ext NCR, Int NCR, RFI, and RMI for the same project-month.
PQI - Variance	Auto-Calculation (Read-Only)	= [PQI Acc. till this Month] - [PQI Acc. till Last Month].
PQI - This Month Score	Auto-Calculation (Read-Only)	= Average of (Closure Rate this Month) across Ext NCR, Int NCR, RFI, and RMI.
Average of All Projects	Auto-Calculation (Read-Only)	= Average of (PQI - Acc. till this Month) across all active projects in the same reporting period.

6.1.9 Material Inspection

Field Name	UI Input Type	Validation / Calculation Logic
Acc. MIRN till Last Month	Auto-Calculation (Read-Only)	Auto populated from previous month's submitted Acc. till this Month value. Defaults to 0 if no prior record exists.
Closed till Last Month	Auto-Calculation (Read-Only)	Auto populated from previous month's submitted Closed till this Month value. Defaults to 0 if no prior record exists.
Issued this Month	Numerical Input	Integer >= 0. Mandatory.
Closed this Month	Numerical Input	Integer >= 0. Mandatory.
Acc. till this Month	Auto-Calculation (Read-Only)	= [Acc. till Last Month] + [Issued this Month]
Closed till this Month	Auto-Calculation (Read-Only)	= [Closed till Last Month] + [Closed this Month]
Closure Rate this Month	Auto-Calculation (Read-Only)	= ([Closed this Month] / [Issued this Month]) x 100.
Variance	Auto-Calculation (Read-Only)	= [% Acc. till this Month] - [% Acc. till Last Month]. Positive = improvement. Display with +/- indicator and colour coding.
OSD Available	Radio Button	Yes, No
No. of MIRNs with OSD	Numerical Input	Integer >= 0. Mandatory if OSD available is Yes
Total Overage Quantity	Numerical Input	Integer >= 0. Optional
Total Shortage Qty	Numerical Input	Integer >= 0. Optional
Total Damage Qty	Numerical Input	Integer >= 0. Optional
Total Defective Qty	Numerical Input	Integer >= 0. Optional
Total No. of Material/Items in MIRN	Numerical Input	Integer >= 0. Optional
Approved	Numerical Input	Integer >= 0. Enabled if Total No. of Material/Items in MIRN >0
On Hold Quarantine	Numerical Input	Integer >= 0. Enabled if Total No. of Material/Items in MIRN >0
Rejected Do not use	Numerical Input	Integer >= 0. Enabled if Total No. of Material/Items in MIRN >0
Rejected – Return to Supplier	Numerical Input	Integer >= 0. Enabled if Total No. of Material/Items in MIRN >0
Hazardous Material	Numerical Input	Integer >= 0. Enabled if Total No. of Material/Items in MIRN >0
Handle & Store with care	Numerical Input	Integer >= 0. Enabled if Total No. of Material/Items in MIRN >0

6.1.10 QTBT

Field Name	UI Input Type	Validation / Calculation Logic
Acc. till Last Month	Pre-filled (Read-Only)	Auto populated from previous month's submitted Acc. till this Month value. Defaults to 0 if no prior record exists.
Manhours till last month	Pre-filled (Read-Only)	Auto populated from previous month's submitted Closed till this Month value. Defaults to 0 if no prior record exists.
Total Nos of QTBT this Month	Numerical Input	Integer >= 0. Mandatory.
No of attendees	Numerical Input	Integer >= 0. Mandatory if Total Nos of QTBT this Month > 0
Duration(minutes)	Numerical Input	Integer >= 0. Mandatory if Total Nos of QTBT this Month > 0
Total Manhours this month	Auto-Calculation (Read-Only)	= [Acc. till Last Month] + [Issued this Month]
Acc QTBT till this Month	Auto-Calculation (Read-Only)	= Acc till Last Month + Total Nos of QTBT this Month
Acc Manhours till this Month	Auto-Calculation (Read-Only)	= Manhours till Last Month + Total Manhours this Month

6.1.11 Project Monthly Document Status – Submittals & Drawings

Field Name	UI Input Type	Validation / Calculation Logic
Document Type	Static Label	Drawings, Submittals
Approved	Numeric Text	Integer >=0 Mandatory
Re-submitted	Numeric Text	Integer >=0 Mandatory
Rejected	Numeric Text	Integer >=0 Mandatory
Under Review	Numeric Text	Integer >=0 Mandatory
Remarks	Multi Line Text	Optional

Review Time:

Average Client Review Time	Numeric Text	Integer >=0 Mandatory
Average Review Time (Aljihaz)	Numeric Text	Integer >=0 Mandatory

6.1.12 QMS Report

Field Name	UI Input Type	Validation / Calculation Logic
Department (per row)	Dropdown	HSSE, Quality, Finance, Fleet & Facilities Management, Human Resources, Group Digital & Technology, Operations
Type (per row)	Dropdown	Manual, Policy, SOP & Form
Document Name	Text field	Optional
Status (per row)	Dropdown	Under Review and Signature, Approved & Published
Remarks	Multi Line Text	Optional

6.1.13 Quality Assessment Brief

A free-text field capturing the QAQC representative's qualitative assessment of project quality performance for the month.

Field Name	UI Input Type	Validation / Calculation Logic
Quality Assessment Brief	Multi-line Text Area	Manual entry. Mandatory. AI Suggestion feature available - see below. Final text reviewed and confirmed by QAQC representative before submission.

AI SUGGESTION— Quality Assessment Brief - Auto-Summary Generator

The AI will analyse and summarize:

(1) Month-over-month variance for each category and to escalate if PQI score is below the target for 3 consecutive months (Ext NCR, Int NCR, RFI, RMI, PQI, Mat).

(2) Categories with negative variance

(3) Any open critical NCRs exceeding 45 days.

(4) The top-performing and lowest-performing metric category with actionable recommendations.

The QAQC representative may accept, edit, or entirely override the AI-generated text before submission. All AI interactions are logged for audit purposes.

6.2 Customer Satisfaction Module

Field Name	UI Input Type	Validation / Calculation Logic
Project Name	Auto-fill (Read-Only)	Dropdown from the project master
Overall quality of executed work	Radio Button (1–5 scale)	Mandatory. Scale: 1 = Very Dissatisfied, 2 = Fair, 3 = Good, 4 = Very Good, 5 = Excellent.
Project completed within the scheduled timeline	Radio Button (1–5 scale)	Mandatory. Scale: 1 = Very Dissatisfied, 2 = Fair, 3 = Good, 4 = Very Good, 5 = Excellent.
Communication throughout the project	Radio Button (1–5 scale)	Mandatory. Scale: 1 = Very Dissatisfied, 2 = Fair, 3 = Good, 4 = Very Good, 5 = Excellent.
Professionalism and courtesy of our team	Radio Button (1–5 scale)	Mandatory. Scale: 1 = Very Dissatisfied, 2 = Fair, 3 = Good, 4 = Very Good, 5 = Excellent.
Value for money of the project	Radio Button (1–5 scale)	Mandatory. Scale: 1 = Very Dissatisfied, 2 = Fair, 3 = Good, 4 = Very Good, 5 = Excellent.
How well did we handle any issues or concerns that arose?	Radio Button (1–5 scale)	Mandatory. Scale: 1 = Very Dissatisfied, 2 = Fair, 3 = Good, 4 = Very Good, 5 = Excellent.
Most Satisfactory Aspects	Multi-line Text Area	Optional free text.
Improvement Suggestions	Multi-line Text Area	Optional free text.
Project Completed as per Expectations	Radio Button (Yes / No / Partially)	Mandatory.
Would Recommend Our Services	Radio Button (Yes / No / Partially)	Mandatory.
Additional Comments / Recommendations	Multi-line Text Area	Optional free text.

6.3 Daily Document Submission Report

6.3.1 Project Details

Field Name	UI Input Type	Validation / Calculation Logic
Project Name	Dropdown	Pulled from active Project Master list. Mandatory.

Field Name	UI Input Type	Validation / Calculation Logic
Report Date	Date Picker	Mandatory
PM Name	Auto filled	Fetches from active Project Master list.
PE Name	Auto filled	Fetches from active Project Master list
DC Name	Auto filled	Fetches from active Project Master list.

6.3.2 Daily Submission for Submittals & Drawings (for each metric)

Field Name	UI Input Type	Validation / Calculation Logic
No Submissions	Toggle Button	Yes/No
Discipline	Static Text	The UI must render 5 distinct discipline blocks (Civil, Mechanical, Structural, Electrical, Instrumentation) under the Drawings category, and 5 distinct discipline blocks under the Submittals category.
Metric	Static Text	Approved, Resubmit, Rejected, Under Review & Total
Yesterday	Static Text	Auto populated from previous day's submission for each metric
Current Date	Numeric Text	Integer >=0. Editable input. Mandatory for all metrics except Total
Total Submission	Auto filled	Read-only for both Yesterday and Current Date columns. Auto-calculates in real-time as: Sum of Approved + Resubmit + Rejected + Under Review.

6.3.3 Approved Status by Revision for Submittals & Drawings (for each revision)

Field Name	UI Input Type	Validation / Calculation Logic
Revision No.	Auto filled	Increments automatically with each new revision
Yesterday	Static Text	Auto populated from previous day's submission for each metric
Current Date	Numeric Text	Integer >=0 Mandatory (except for Total)

6.3.4 Submissions by Document type

Field Name	UI Input Type	Validation / Calculation Logic
Document Type	Static Text	Drawings Material Submittals Method Statements IPTs CVs PQP IFR Vendors
Yesterday	Static Text	Auto populated from previous day's submission for each metric
Current Date	Numeric Text	Integer >=0 Mandatory (except for Total)

6.3.5 Submission Review – Pending Days

The below details just capture every day's snapshot values for each status with respect to the entities.

Field Name	UI Input Type	Validation / Calculation Logic
Entity	Static Label	Client, Alghaz, Supplier
Status Under Review	Numeric Text Data is entered against <=7 days 8-30 days >=31 days	Integer>=0 Mandatory
Status A	Numeric Text Data is entered against <=7 days 8-30 days >=31 days	Integer>=0 Mandatory
Status B	Numeric Text Data is entered against <=7 days 8-30 days >=31 days	Integer>=0 Mandatory
Status C	Numeric Text Data is entered against <=7 days 8-30 days >=31 days	Integer>=0 Mandatory
Status D	Numeric Text Data is entered against <=7 days 8-30 days >=31 days	Integer>=0 Mandatory
Status E	Numeric Text Data is entered against <=7 days 8-30 days >=31 days	Integer>=0 Mandatory
Total	Auto filled	Summation of values entered in all status

6.3.6 Correspondence

Field Name	UI Input Type	Validation / Calculation Logic
Correspondence	Static Text	Outgoing, Incoming
Yesterday	Static Text	Auto populated from previous day's submission for each metric
Current Date	Numeric Text	Integer >=0 Mandatory (except for Total)

7. Escalation Matrix

Escalations are triggered automatically if the closure rate for each category is below the set target consecutively for two months. This escalation matrix currently applies to the QA/QC Report module.

Level	Risk Combination
L1	Triggered to the Quality member of the project when the Closure rate for the month is below the defined target consecutively for two months
L2	Escalated to Project Head if the closure rate is less than the target consecutively for three months

L3	Escalated to BU head/department head if the closure rate is less than the target consecutively for four months.
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In addition to above escalation matrix, each module submission also has an escalation which will be sent if the form remains unapproved by the approver.

Level	Risk Combination
P2	First reminder triggered to the Current Pending Approver (Corporate QC Manager, Data Governance Manager) if the report remains unapproved for 2 days, repeating daily until it reaches P1.
P1	Triggered to the QAQC Director if it remains unapproved for 5 days, repeating daily until the report gets approved.

In addition to approval delays, the system will actively monitor for missing submissions and trigger automated alerts to ensure timely data entry.

Alert Type	Risk Combination
Daily Submission Alert	- If the Document Controller fails to submit the daily report by 11:00 AM the following day, an automated alert is sent to the Data Governance Manager at 11:01 AM. - Triggered to the QAQC Director at 11:01 AM if it remains unsubmitted by 11:00 AM the following day, repeating daily until the report gets submitted..
Monthly Submission Alert	- If the QAQC Representative fails to submit the monthly report by the 7th of the current month, an automated alert is sent to the Corporate Quality Manager on the 8th at 11:00 AM. - Triggered to the QAQC Director at 11:01 AM if it remains unsubmitted for 2 days, repeating daily until the report gets submitted.

8. Functional Requirements as User Stories

8.1 QAQC Metrics Module

ID	Field	Requirement	Acceptance Criteria
FR-1	Project Details	As a QAQC representative, I want to select a project and reporting month, so that I can enter the data for the monthly report.	- Project dropdown populated from active Project Master. - Month/Year picker restricts selection of future month - System prevents duplicate submissions. - Cost Centre gets auto filled. - The user must strictly block the system from initiating data entry for the current month if the previous month's report has not yet been submitted.
FR-2	PQP Status	- As a QAQC representative, I want to select the current PQP status from a predefined list so that I can accurately reflect the project's quality planning phase. - As a QAQC representative, I want to see the approval date field dynamically appear only when an "Approved" status is selected so that the form remains uncluttered.	- The dropdown must contain exactly the requested statuses (Approved A/B/C, under preparation, Under Review with client, Rejected, Others) and reveal a text box if 'Others' is chosen. - The "PQP Date of Approval" field must remain entirely hidden unless the user selects

			"Approved A", "Approved B", or "Approved C".
FR-3	Report Reference	- As a QAQC representative, I want to log the reference number of the previous client report so that I can maintain a clear chain of documentation. - As a QAQC representative, I want to specify the exact calendar date range for the current report so that I can establish clear boundaries for the reported data.	- The "Last Report Sent to Client" must accept standard alphanumeric text inputs to accommodate various tracking number formats. "Report Date (From)" and "Report Date (To)" must strictly accept valid date formats via a standard UI date-picker widget only when 'Last Report Sent to Client' is filled.
FR-4	Meetings Conducted	- As a QAQC representative, I want to add multiple dynamic rows for different meeting types so that I can comprehensively log all meetings. - As a QAQC representative, I want to log both the last conducted date and the next scheduled date so that I can ensure quality meetings are occurring at the correct cadence.	- The system must provide a "+ Add Row" button that appends a new entry row with Meeting Type, Last Review Date, and Next Review Date fields. - The "Meeting" dropdown must restrict inputs to: Project Management Review Meeting, Internal Meeting, External Meeting, and Project Quality Meeting.
FR-5	QMS Internal Audit	- As a QAQC representative, I want to mention if an internal audit was conducted this month so that I can comply with QMS reporting standards. - As a QAQC representative, I want to log the specific dates of past and future audits so that I can properly map out compliance schedules.	"Is Internal Audit Conducted?" must be a mandatory Yes/No dropdown field. "Last Internal Audit Conducted Date" and "Next Internal Audit Conducted Date" must utilize standard calendar date pickers.
FR-6	Quality Manpower	- As a user, I want to list manpower by department so that I can showcase the resource allocation on my site. - As a user, I want to have the approval/rejection fields automatically disabled if client approval is not required so that I am prevented from entering unnecessary zero-values.	- If "Is client approval required?" is set to "No", the "Approved" and "Rejected" input fields must immediately disable and clear their values. - The system must dynamically calculate and display the total sum of Manpower Nos, Approved, and Rejected fields in a read-only footer row.
FR-7	Historical Autofill	As a QAQC representative, I want last month's accumulated and closed numbers auto populated, so that I do not need to manually re-enter historical data.	- Acc. till Last Month and Closed till Last Month populated from previous submitted record. Fields are read-only. - If no prior record, defaults to 0 with visible notification.
FR-8	Current Month Entry	As a QAQC representative, I want to enter Issued this Month and Closed this Month numbers for each category, so that the system can compute all performance metrics.	Numerical input fields with integer ≥ 0 validation. System calculates all other fields on entry.
FR-9	Auto-Calculation	As a QAQC representative, I want all cumulative totals, closure rates, and variances computed	All auto-calculated fields are read-only. Formulae match Section 7B exactly.

		automatically, so that I do not want to manually calculate.	Special rule applied: Closure Rate = 100% when Issued = 0 and Closed = 0 for active project.
FR-10	Open NCR Details	- As a QAQC representative, I want the system to calculate the Open NCR. - Then I can assign Open NCRs to specific Business Units (BUs) and ageing buckets so that I can easily identify departmental bottlenecks.	- Open NCRs" must be automatically computed in real-time as (Acc. till Last Month + Issued this Month) - (Closed till Last Month + Closed this Month). - The system must block final submission and throw an error if the sum of the counts inside the "Open NCR Details" table does not exactly match the auto-calculated total Open NCRs.
FR-11	PQI Calculation	As a QAQC representative, I want the PQI score automatically computed from the four metric categories, so that I have a single quality index.	All auto-calculated fields are read-only. Formulae match Section 6.1.8 exactly.
FR-12	Material Inspection	- As a QAQC representative, I want to input the number of MIRNs issued and closed this month so that the system can automatically calculate my cumulative totals, closure rate, and variance. - As a QAQC representative, I want to specify if OSD (Overage, Shortage, Damage) is available so that I can log the specific defect quantities and the number of MIRNs affected.	- The system must display read-only fields for previous month data and automatically calculate "Acc. till this Month", "Acc. Closed till this Month", "Closure Rate this Month", and "Variance" based on the "Issued this Month" and "Closed this Month" inputs. - Selecting "Yes" for "OSD Available?" must reveal a sub-section to input "No. of MIRNs with OSD", "Total Overage Qty", "Total Shortage Qty", "Total Damage Qty", and "Total Defective Qty".
FR-13	Status Tag	- As a QAQC representative, I want to categorize inspected materials into specific status tags so that I have a clear record of material status. - As a QAQC representative, I want the system to validate that my distributed status tags match the total items in the MIRN so that I do not accidentally miscount site materials.	- The interface must provide specific input fields for: "Approved", "On Hold Quarantine", "Rejected Do not use", "Rejected – Return to Supplier", "Hazardous Material", and "Handle & Store with care". - The system must enforce a validation rule where the sum of the quantities entered in the Status Tags distribution exactly equals the mandatory value provided in the "Total No. of Material/Items in MIRN" field.
FR-14	QTBT	As a QAQC representative, I want to enter the number of QTBTs conducted, attendees, and duration so that I can log this month's safety and quality training efforts.	The system must enforce "Total Nos of QTBT this Month", "No of Attendees", and "Duration (minutes)" as mandatory input fields.

		As a QAQC representative, I want the system to automatically compute total manhours and accumulated totals so that I do not have to perform manual calculations.	The system must automatically calculate and display "Total Manhours this Month", "Acc QTBT till this Month", and "Acc Manhours till this Month" based on the current month's inputs and the read-only historical data ("Acc till Last Month" and "Manhours till Last Month").
FR-15	Project Monthly Document Status – Submittals & Drawings	As a Document controller, I want to enter the number of submittals and drawings against the status and also the review time taken by Aljihaz & Client so that I can understand how many documents need attention.	The system must get the input as integer ≥ 0 and get remarks as text.
FR-16	QMS Report	As a Document controller, I want to enter a department, type, document name and its status so that I can capture monthly status of QMS reports.	- The dropdown must contain HSSE departments, Quality, Finance, Fleet & Facilities Management, Human Resources, Group Digital & Technology and Operations for department. - The dropdown must contain Manual, Policy, SOP & Form for type - The dropdown must contain Under Review & Signature, Approved & Published
FR-17	Quality Assessment Brief	As a QAQC representative, I want to enter a manual Quality Assessment Brief and receive an AI-generated draft to review, so that I can produce an effective brief on all the above quality metrics of projects.	- Text area provided for manual entry. AI Suggestion generates draft based on current data. AI draft is editable and not mandatory. - Final text saved with submission.

8.2 Customer Satisfaction Module

FR-18	Project Selection	As a QAQC representative, I want to select a project, so that I can enter the data for the Customer Satisfaction Survey.	Project dropdown populated from active Project Master.
FR-19	CSAT Survey Entry	As a CSAT representative, I want to rate service parameters and submit qualitative feedback for a project, so that customer satisfaction is captured in a structured format.	Six rating questions (1–5) and two Yes/No/Partially questions are mandatory. Survey can be saved as draft or submitted for approval.

8.3 Daily Document Governance Module

FR-20	Project Details & Report Period	- As a Document Controller, I want to select my project from a dropdown list so that I am logging document metrics against the correct project database. - As a Document Controller, I want to define the exact dates of the submission so that project	The system must strictly block the Document Controller from selecting a date or entering data if the immediately preceding calendar day's report has not yet been submitted.
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		stakeholders understand the timeframe this data covers.	- The "Project Name" field must be a mandatory dropdown; submission must be blocked if left blank. - The app must utilize standard date pickers. - System should not allow the DC to enter the data for date without entering details for the previous date.
FR-21	PM Name, PE Name, DC Name	As a Document Controller, I want the system to automatically retrieve and display the Project Manager, Project Engineer, and DC names based on the selected project so that I do not have to type them manually every time.	- Upon selecting a valid "Project Name," the PM, PE, and DC fields must instantly populate with predefined master data. - These three fields must be visually locked to prevent manual overriding by the user.
FR-22	"No Updates" / Zero-Fill Toggle	As a Document controller, I want a quick toggle to indicate that there is no new activity for the current reporting date, so that I can instantly update the form with 0 for all fields and submit it without manual typing	- A 'No Submissions' toggle must be available at the top of the data entry form." "When activated, the system automatically fills all editable numerical 'Current Date/Month' fields with '0'." "The user can immediately click Submit, allowing them to clear their pending task in seconds."
FR-23	Daily Submission Discipline Blocks for Submittals & Drawings	- As a Document Controller, I want to enter current date data into 5 distinct discipline blocks for both drawings and submittals. I want the system to automatically load my previous date values and auto-calculate the Totals. - As a Document Controller, I want the system to automatically load my previous date values.	- The UI must display only two columns for the metrics: 'Yesterday' (read-only) and 'Current Date' (editable input) - The 'Total' row must be strictly read-only for both columns and auto-calculate as Total = Approved + Resubmit + Rejected + Under Review
FR-24	Approved Status by Revision	- As a Document Controller, I want to dynamically add rows for specific revisions and add current approved nos (e.g., Rev 00, Rev 01) so that I can track exactly which version of the documents are being approved. - As a Document Controller, I want the system to track submittal revisions and drawing revisions in two distinct, separate tables so that the data remains organized and easy to read.	- The UI must provide separate "+ Add Revision" buttons for the Submittals section and the Drawings section. - Clicking the add button must generate a new row where the user can input the Revision Name (e.g., "Rev 03") and the Current Date value.
FR-25	Submission by Document Type	As a Document Controller, I want to input the number of submissions for the selected date with respect to the document type so that the DC can track the type of documents submitted.	- The UI must display only two columns for the metrics: 'Yesterday' (read-only) and 'Current Date' (editable input) - The yesterday column fields must be read-only and pre-populated based on the previous submitted report for

			that project (or default to 0 if no history exists). Current Date field must be an integer and entered by the user
FR-26	Submissions Review - Pending Days	- As a document controller, I want a consolidated grid showing pending documents categorized by Entity, Status, Ageing and one total column per entity row so that I can easily identify bottlenecks - As a document controller, I want a structured matrix input for pending days so that I don't have to create separate manual trackers for Client, Aljihaz, and Supplier delays.	- The table must contain three fixed rows for entities: Client, Aljihaz, and Supplier. - The table headers must accurately group the 6 specific statuses (Under Review, Status A, Status B, Status C, Status D, Status E). - Under each status, there must be three editable numerical input columns for the ageing buckets: ≤7, 8-30, and ≥31.
FR-27	Correspondence	As a Document Controller, I want to input the number of incoming and outgoing letters for the selected date so that the system can automatically update the project's cumulative correspondence tracker.	- The UI must display only two columns for the metrics: 'Yesterday' (read-only) and 'Current Date' (editable input) - The yesterday column fields must be read-only and pre-populated based on the previous submitted report for that project (or default to 0 if no history exists). - Current Date field must be an integer and entered by the user

8.4 Common Functional Requirements

FR-28	Submission & Approval	- As a QAQC representative & Document controller, I want to submit the completed report for approval, so that data is validated before distribution. - As an approver I want to approve the entered details so that I can verify if the data is correct.	- Submit button available when all mandatory fields completed. - Status: Draft > Submitted > Approved / Sent Back.
FR-29	Report Distribution	As the system, I want to distribute the approved report to designated recipients, so that project heads receive timely information on quality metrics.	- The system shall automatically generate and distribute an email containing the report and dashboard layouts to project stakeholders on a fortnightly basis for document status and monthly basis for Monthly Status. - It should NOT send immediately upon every individual verification.
FR-30	Master Configuration	As an Admin, I want to configure project master details including targets, so that the system can enforce benchmarks and auto-populate fields accurately.	- Admin interface for Project Master. - Target percentages configurable per project per KPI.

FR-31	Dashboard	- As a user, I want a real-time dashboard with all project quality and QMS KPIs, so that I have instant visibility into quality performance. (Dashboard design would be provided by Digital Team) - Main Dashboard - Dashboard layout for distribution	- Dashboard displays KPIs across all projects. Filter by Project group, Project Name, Month, Category. Drill-down to project-level detail.(Dashboard final layout will be provided by Digital team) Default View: Upon loading, the daily document dashboard shall default to the maximum (latest) available date in the database, displaying the absolute cumulative snapshot for that date (e.g., total active items, current backlog) with no delta calculations. Filtered View: When a user selects a specific past date or a custom "From / To" date range, the system shall execute an automated subtraction algorithm (End Date value MINUS Start Date value) to display the exact net movement (FTD deltas) that occurred during that specific timeframe.
FR-32	Data Export	As a user, I want to export report data & dashboard in PDF and Excel formats, so that I can use data for presentations and management reviews.	Export option available on all report views. PDF replicates approved dashboard and report layout. Excel provides raw data table with all fields. (Report layout would be provided by Digital team)
FR-33	Escalation	As a System Admin, I want the system to automatically track the "Days Unapproved" for every submitted report so that the logic for P1, P2, and P3 escalations can be executed.	- For every report transitioning to the Submitted state, the system must capture a Submission Timestamp. The system must calculate and store a real-time Days Unapproved integer representing the calendar days elapsed since that timestamp. - The calculated Days Unapproved integer must serve as the primary trigger variable for the P1 and P2 automated escalation protocols defined in Section 7. - The system must also run scheduled chron-jobs to track calendar dates and trigger the Daily (11:01 AM next day) and Monthly (8th day at 11:00 AM) Submission Alerts defined in Section 7
FR-34	Excel Template Upload	As a QAQC Representative & Document Controller, I want to download a standardized Excel template and upload my data, so that I can perform easy data entry instead of manually typing into the web form.	- System provides a clearly formatted, downloadable Excel template for the respective modules. - System validates the uploaded file for errors (e.g., missing

			mandatory fields, wrong data types) before saving. Successfully uploaded data automatically populates the front-end web form for review prior to final submission
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9. Configuration & Masters

The system shall be designed for easy adaptation to new projects & departments.

Master / Config	Setup Type	Remarks
Project Master	One-time setup	Project Group, Project Name, Project Code, Location, Cost Centre, Target KPI %, QAQC Representative, Document Controller, Project Manager, Approval Hierarchy, Distribution List.
User Master	One-time setup	Name, Role (RBAC), Email, SBG, BU, Project name, Mobile number List of users to be provided by Digital Team.
Target Benchmarks	Configurable per project	Target % for each KPI category. Editable by Admin.
Distribution List	Configurable per project	Email recipients for approved report PDF distribution.
Report Template	One-time setup	Report PDF template as approved by Digital Team. Report Dashboard template as approved by Digital Team Versioned and locked post-approval.

10. Reports & Dashboards

S.No	Report / Dashboard Name
1	Monthly Project Quality Report (per project, per month) – Dashboard Report and Report as PDF(sent in email)
2	Biweekly Document Governance Report(per project fortnight basis) - Dashboard Report and Report as PDF(sent in email)
3	Daily Document Governance Log
4	CSAT Summary & Trend Report (per project and consolidated across projects)
5	All-Project Quality Summary Dashboard (consolidated dashboard integrating data from all modules) - Main Dashboard

11. Role-Based Access Control (RBAC)

The following roles apply uniformly across all three modules (QA/QC Report, CSAT, Daily Document Governance Log, with module- and record-level permissions applied within each workspace.

Role	Phase / Access	Permissions & Responsibilities
QAQC Representative	Data Entry & Submission	Enter current month data and submit report for approval.
Approver / Reviewer	Approval Phase	Review submitted reports, approve or send back with comments.
Quality Manager	All Phases	View all submitted and approved reports, access trend and dashboards, export reports.
Project Head	Post-Approval	Receive report PDF, view project-level KPI dashboard in portal. Receives escalation email.

Role	Phase / Access	Permissions & Responsibilities
Admin	All Phases	Responsible for the governance, Manage user roles, escalation matrix.
Document Controller	Document Management – Data Entry	Create/edit document register entries, log revisions, and capture submissions of submittal & drawing until the register is locked by submission or approval.

12. Assumptions

- Master data (Projects, Users, etc.) will be provided by Digital team as and when new project comes.
- Main Dashboard, Dashboard Report and Report for distribution will be provided by the Digital Team; the development team will replicate the approved dashboard.
- Mobile application will be developed for Android and iOS
- Phase 1 will operate with fully manual entry until NCR/RFI/RMI modules are available
- The development team and digital team will be available for requirement clarifications, reviews, and timely approvals throughout the project lifecycle.
- Historical data to be migrated from prior periods (if any) will be provided in a structured format by the Digital Team

13. Sign-Off & Approvals

This document requires formal review and approval from the Digital Team and Quality team before development and deployment.

Prepared By

Name	Department	Signature	Date
Subbu Sankar V R	Digital Team	V.R. Subbu Sankar	18-08-2026

Approved By (Quality Team)

Name	Department	Signature	Date
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